

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

AI Career Coach And Resume Optimization

Team details :

Aadil Verma (2210991102)

Aryan Thakur (2210991391)

Harshit Kumar (2210991635)

Vansh Thakur (2210992499)

Supervised By: Dr. Rajat Takkar

Department of Computer Science and Engineering

Chitkara University, Punjab

Table of Contents

- Introduction
- Problem Statement
- Technical Details
- System Design
- Objectives
- Project Highlights
- Key Features
- Conclusion
- Future Scope
- References/Links used



Fig.1

Introduction

AI Personalized Learning is an intelligent career insights platform designed to help professionals and job seekers stay ahead in their careers by analyzing real-time market trends, salary data, and skill demands

It provides users with valuable insights to make informed career decisions. With a seamless onboarding process, secure login, and AI-powered analytics, users can explore industry growth, demand levels, and top skills required in their field. Whether you're a fresher looking for career guidance or an experienced professional aiming for growth, it empowers you with data-driven recommendations.



Fig.2

Problem Statement

In today's fast-changing job market, many individuals struggle with several critical challenges:

Uncertainty in Career Planning

Lack of clear guidance on the right career path.

Resume & Interview Challenges

Difficulty in crafting strong resumes and preparing effectively for interviews.

Lack of Industry Insights

Limited access to real-time job market trends and growth opportunities.

Skill Gap Identification

No clear understanding of required skills and areas for improvement.

The Solution: AI Career Coach

Addresses these challenges by providing **personalized career guidance**, **AI-driven resume analysis**, **interview preparation**, and **real-time industry insights**, helping users make informed career decisions.



Fig.3

Technical Details

- ◆ **Frontend:** Next.js, ShadCn UI, JavaScript
- ◆ **Backend:** Node.js, Express.js, PostgreSQL
- ◆ **Authentication & Security:** JWT (JSON Web Token) by Clerk
- ◆ **Cloud & Infrastructure:** Hosting (Vercel), Database Service (Supabase, PostgreSQL)
- ◆ **ORM & Database Management:** Prisma ORM
- ◆ **AI Integration:** Gemini APIs (for AI-driven features like interview preparation and resume analysis)
- ◆ **Testing & Monitoring:** Postman (for API testing and middleware validation)

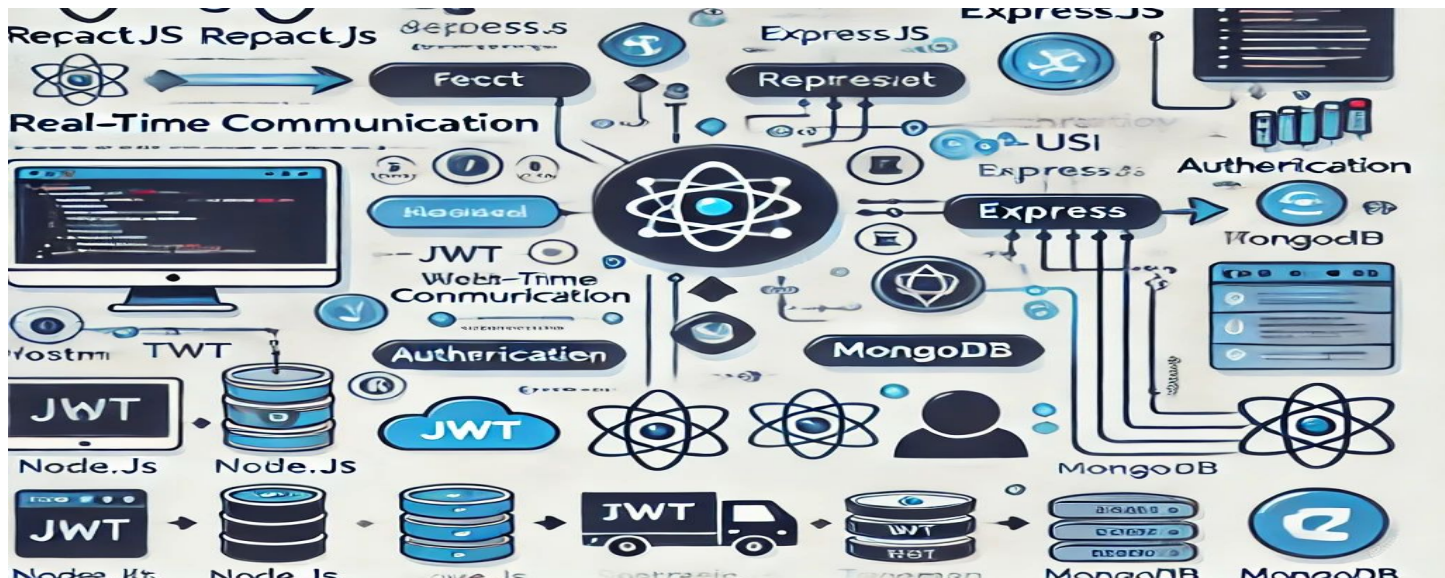


Fig.4

The system design outlines the complete technical structure of the AI Career Coach, detailing how components interact and are managed across the stack.

1 Architecture

The application architecture follows a modular full-stack approach:

- Frontend: Developed using Next.js with Chakra UI and shadcn/ui for responsive UI components
- Backend: Built on Node.js and Express.js, handling routing, authentication, and logic
- Database: PostgreSQL hosted on Supabase, managed using Prisma ORM
- Authentication: Handled securely using Clerk and JWT tokens
- AI Integration: Gemini APIs are used for career suggestions, resume analysis, and interview prep

2 Flow:

User → Next.js → Express API → Clerk (Auth) → PostgreSQL (via Prisma) → Gemini API (AI).

2.1 High-Level Design Diagram

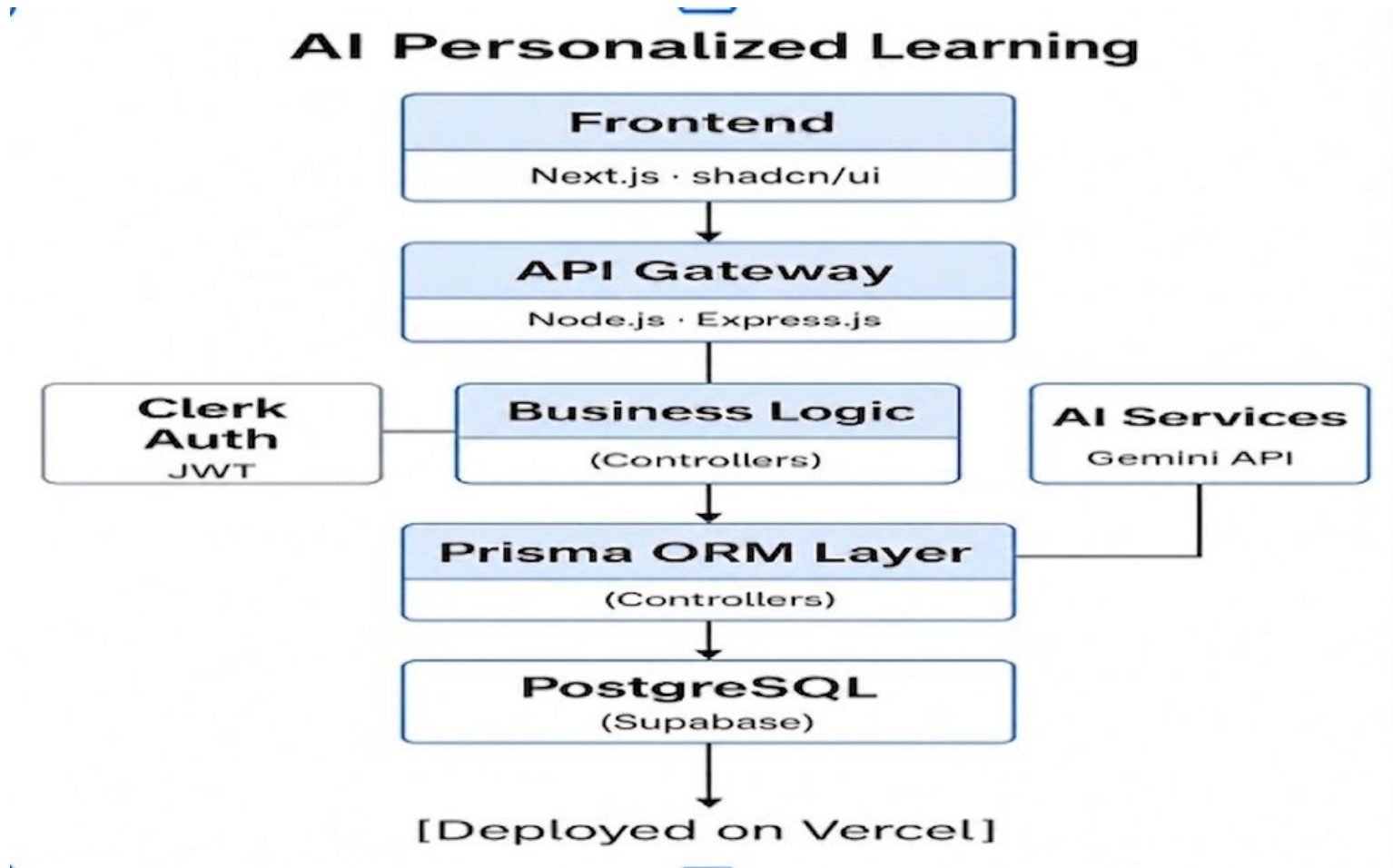


Fig. 5

2.2 ER Diagram

User

- Attributes: user_id, clerkUserId, name, email, bio, industry, timestamps
- Relationships:
 - One user → Many Resumes
 - One user → Many Interviews
 - One user → Many Assessments

Resume

- Attributes: resume_id, user_id (FK), skills, experience, education, filePath

Interview

- Attributes: interview_id, user_id (FK), questions (JSON), score, feedback, date

Assessment

- Attributes: assessment_id, user_id (FK), category, quizScore, improvementTips

ER Diagram:

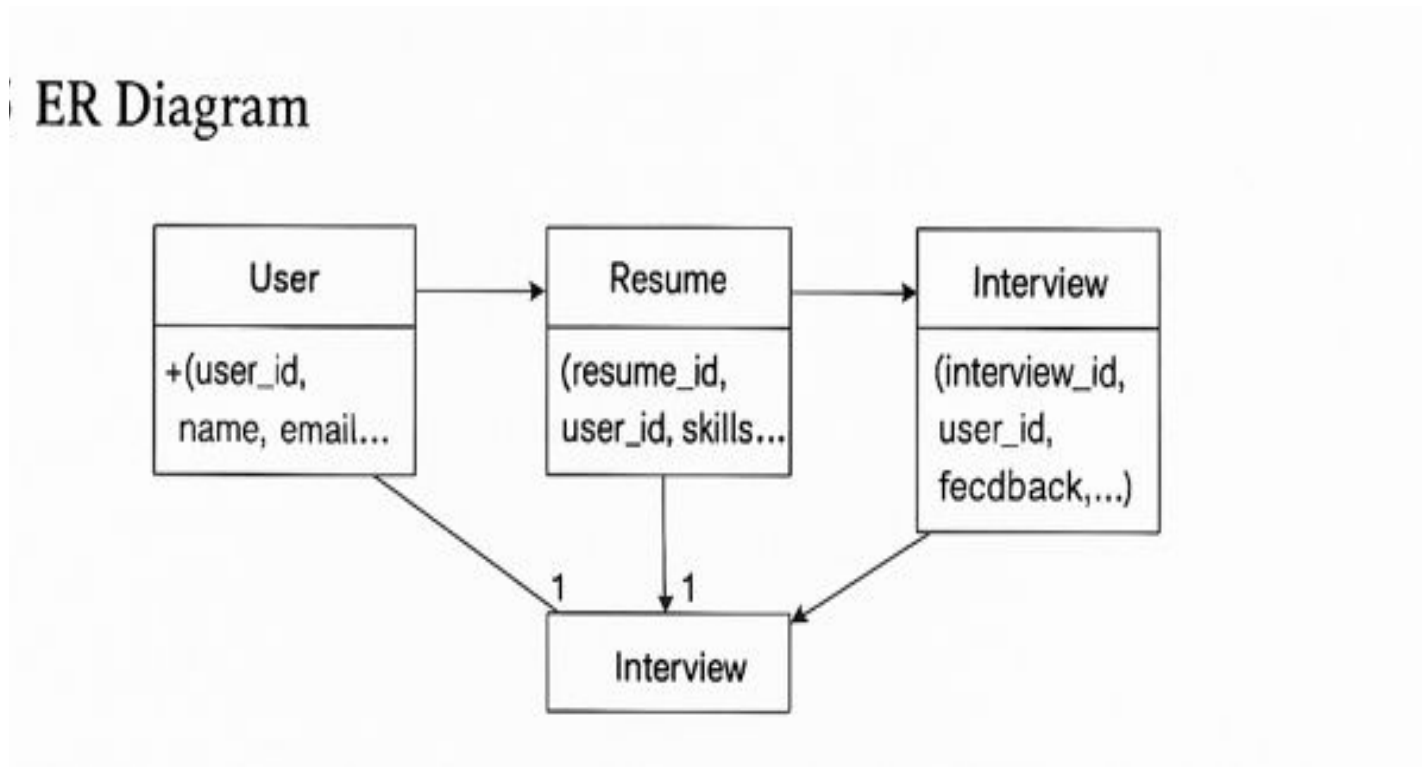


Fig. 6

2.3 UML Diagrams

Class Diagram

- User: userId, name, email, password, industry
- Resume: resumeId, userId, skills, experience
- Job: jobId, title, description, company
- Interview: interviewId, userId, score, feedback

Sequence Diagram – Login Flow

1. User opens login page
2. Sends credentials to /api/auth/login
3. Server validates and returns JWT + Clerk session
4. User redirected to dashboard with session token

Activity Diagram – Resume Generation Flow

1. User uploads/responses input
2. Data sent to backend
3. Gemini API processes request
4. Response saved in database
5. Resume preview/downloaded

UML Diagrams

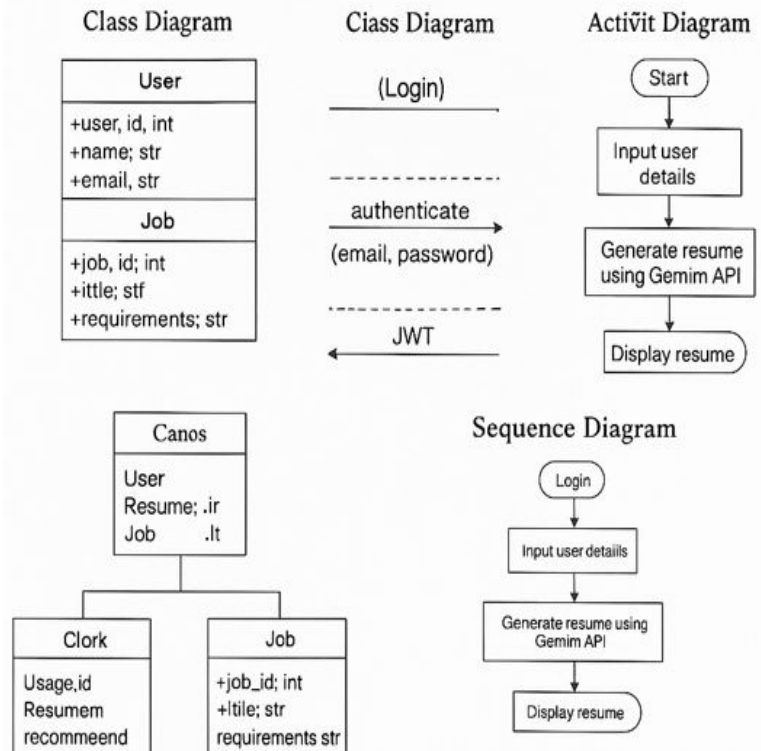


Fig. 7

Objectives

1. **Generate AI-Powered Career Documents:** Create personalized and optimized resumes and cover letters using advanced AI models.
2. **Provide Smart Job Recommendations:** Suggest suitable job opportunities aligned with the user's skills, experience, and career goals.
3. **Enhance Interview Preparation:** Conduct interactive mock interviews and deliver AI-driven feedback with potential Q&A.
4. **Deliver Market Insights:** Offer real-time insights into job trends, in-demand skills, and salary benchmarks across industries.
5. **Track Career Progress:** Maintain a comprehensive history of job applications, interview feedback, and career growth metrics.

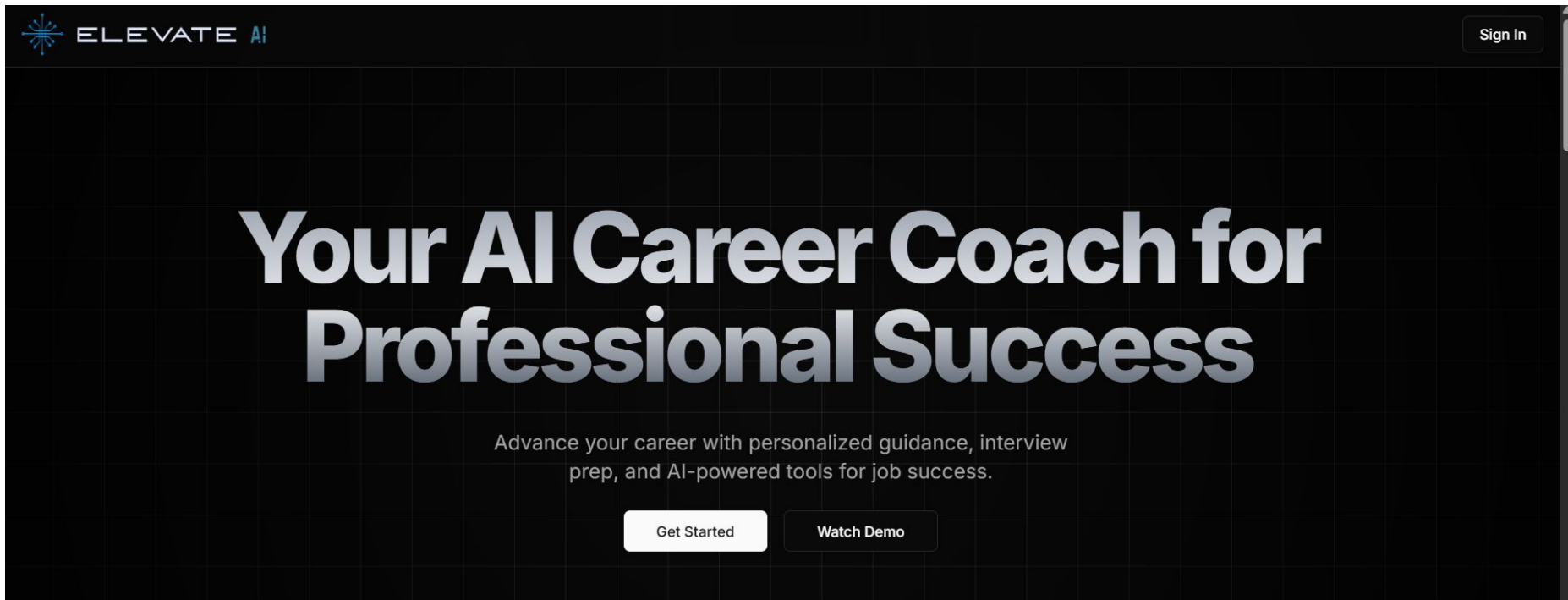
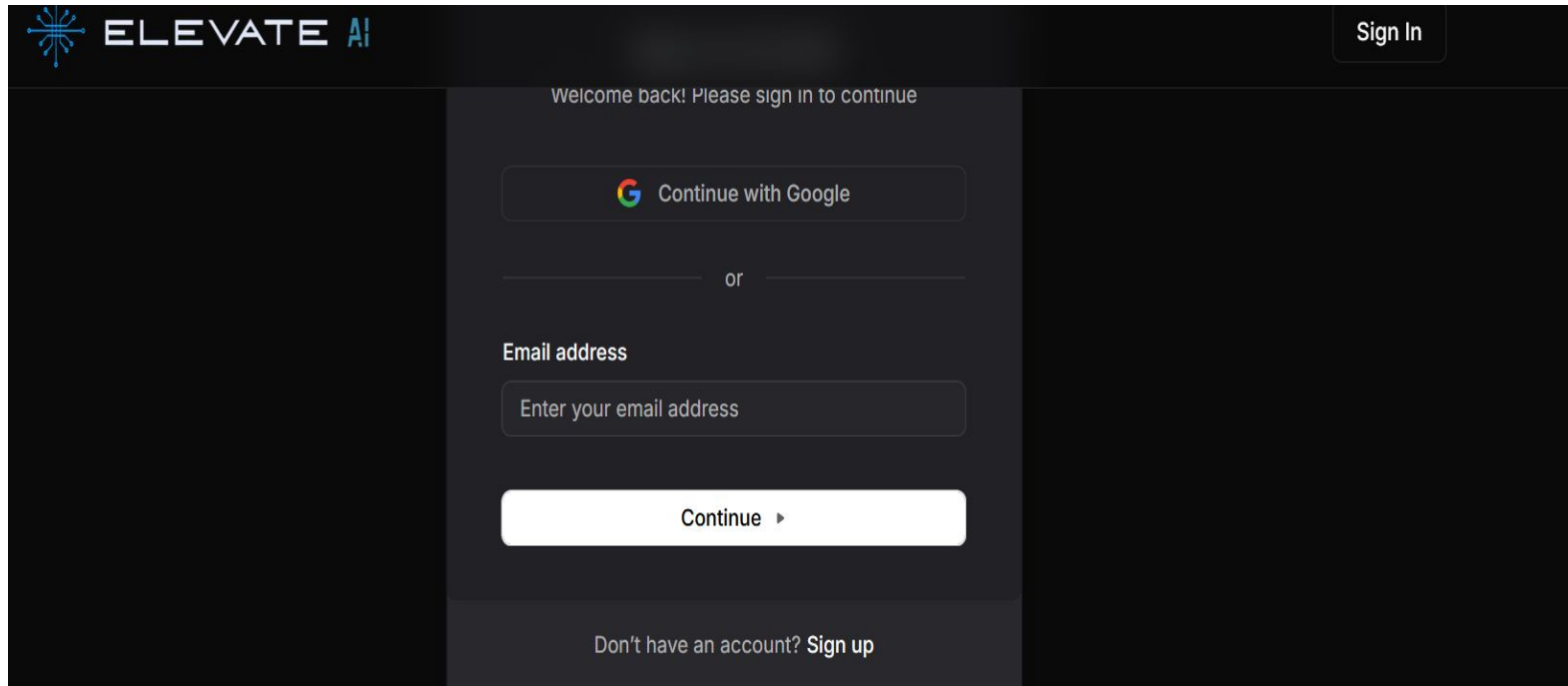


Fig. 8

The Landing Page is the AI Career Coach platform's first impression, designed for engagement. The hero section features a strong headline, subheading, and CTAs for sign-ups, complemented by a career growth image. Four key features are highlighted: AI career guidance, interview prep, industry insights, and smart resume creation. Credibility is built with engagement stats (e.g., 50+ industries, 95% success rate). A smooth user experience is ensured by the top navigation bar.

User Login Page:

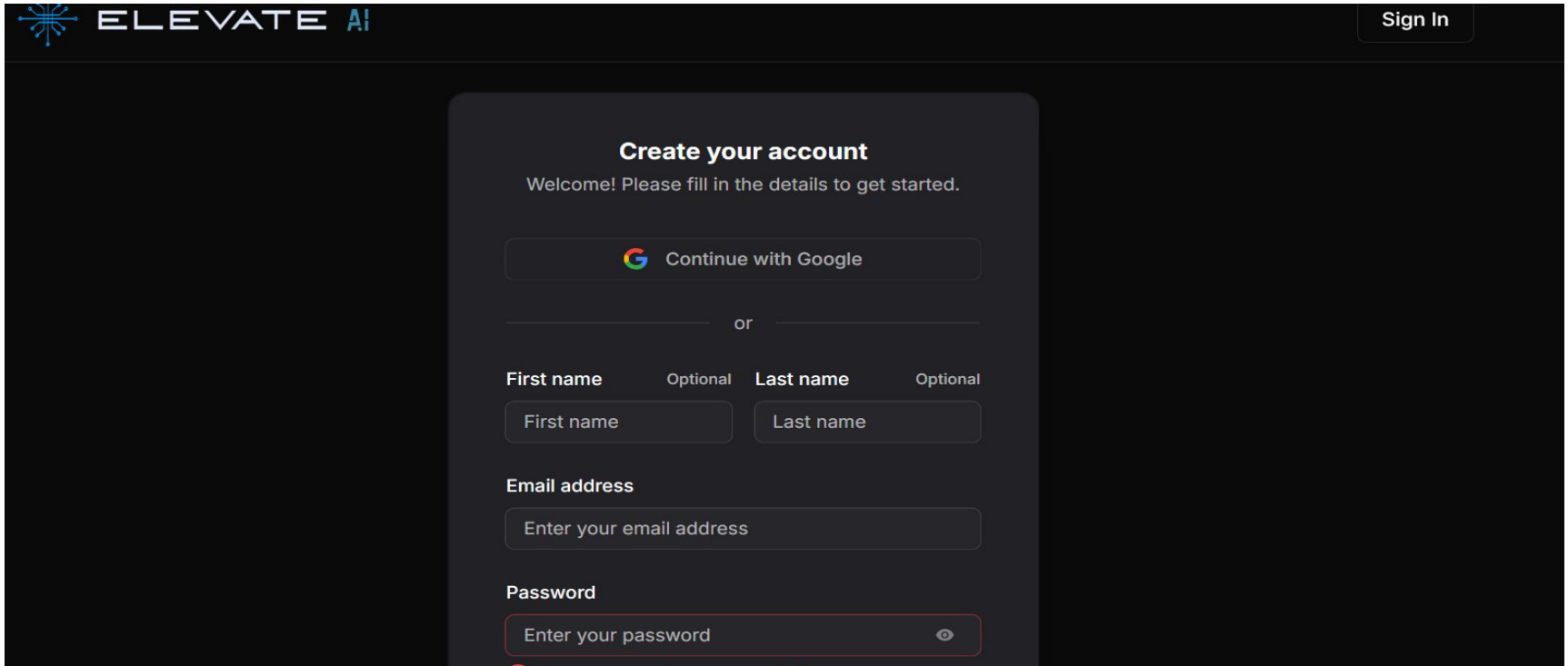


The screenshot displays the login interface for ELEVATE AI. At the top left is the ELEVATE AI logo, and at the top right is a 'Sign In' button. The main content area features a dark background with a central white box containing the following elements: a welcome message 'Welcome back! Please sign in to continue', a 'Continue with Google' button with the Google logo, a horizontal separator with the word 'or' in the center, an 'Email address' label, an input field with the placeholder text 'Enter your email address', and a white 'Continue' button with a right-pointing arrow. At the bottom of the white box, there is a link that reads 'Don't have an account? Sign up'.

Fig. 8

The User Login Page offers secure authentication via email/password or "Sign in with Google" for a quick, private experience. Form validation, Clerk authentication, and JWT secure sessions and enhance user management with a clean, user-friendly design.

User Signup:



The screenshot displays the 'Create your account' page for ELEVATE AI. The page has a dark theme. At the top left is the ELEVATE AI logo, and at the top right is a 'Sign In' button. The main content area is a light gray card with the title 'Create your account' and a welcome message: 'Welcome! Please fill in the details to get started.' Below this is a 'Continue with Google' button. A horizontal line with the word 'or' in the center separates this from the manual registration fields. The manual registration section includes four input fields: 'First name' (with 'Optional' text to its right), 'Last name' (with 'Optional' text to its right), 'Email address' (with a placeholder 'Enter your email address'), and 'Password' (with a placeholder 'Enter your password' and a toggle eye icon). The 'First name' and 'Last name' fields are currently empty, while the 'Email address' and 'Password' fields contain their respective placeholder text.

The **User Sign-Up Page** allows new users to register by providing their first name, last name, email, and password. It also includes a **"Sign Up with Google"** option for quick authentication. The page ensures **data validation** to prevent errors and enforces **secure password storage** to protect user credentials. By offering both manual and Google-based registration, the page enhances **user experience and accessibility**, making the onboarding process smooth and efficient.

Key Features

-  **User Authentication** – Sign up & log in via email/password or Google.
-  **Two-Factor Authentication** – Enhances security for user accounts.
-  **User Onboarding** – Collects industry, skills, and experience details.
-  **Industry Insights** – Displays market trends, demand level, and top skills.
-  **Salary Analytics** – Provides salary ranges based on job roles.
-  **Growth Tools** – Personalized recommendations for career advancement.
-  **Real-Time Updates** – Ensures fresh data for industry insights.
-  **Dark UI Theme** – Modern and professional user interface.
-  **Responsive Design** – Works across desktop and mobile devices.

Future Scope of AI Personalized Learning

As part of **AI Personalized Learning** continuous evolution, we aim to enhance the platform with cutting-edge features that improve user experience, security, and efficiency.

Upcoming Features

AI Mock Interviews & Feedback

- Simulated job interviews with real-time AI evaluation.
- Instant feedback on communication, technical skills, and confidence level.

AI Resume Builder & Skill Verification

Effortlessly create professional resumes backed by verified credentials.

- Provides secure, verifiable skill certificates for recruiters
- Enhances trust and reduces manual validation effort

AI Cover Letter & Job Matching

Generate personalized cover letters and discover job/internship opportunities tailored to your profile.

- AI-curated job suggestions
- Seamless applications with partnered employers
- Boost your chances with role-optimized cover letters

- AI Personalized Learning serves as a powerful, data-driven platform that bridges the gap between career uncertainty and informed decision-making. By combining advanced AI technologies with real-time market insights, it empowers users to identify skill gaps, explore career opportunities, and enhance their professional growth effectively.
- The platform not only simplifies resume building and interview preparation but also provides personalized recommendations tailored to individual goals. With its scalable architecture, secure authentication, and user-friendly design, it ensures a seamless and reliable experience.
- As the job market continues to evolve, AI Personalized Learning has the potential to become an essential tool for students, job seekers, and professionals—guiding them toward smarter career choices and long-term success.

References:

To develop and integrate advanced features in AI Carrier Coach , we referred to the following technologies, tools, and resources:

Authentication & Security

 Clerk Authentication (User Management) – www.clerk.dev

Backend & Database

 Node.js & Express.js for API development – nodejs.org | expressjs.com

 PostgreSQL & Neon for database storage – www.postgresql.org | neon.tech

 Prisma ORM for database management – www.prisma.io

AI & Industry Insights

 Gemini APIs for AI-powered career analysis – ai.google.dev

 Industry Trends & Insights – [LinkedIn Economic Graph](https://www.linkedin.com/pulse/economic-graph-2024-linked-in-research) | Glassdoor Research

Frontend Development

 Next.js Documentation – nextjs.org

 ShadCn UI for UI Components – chakra-ui.com

Testing & Monitoring

 Postman for API testing & middleware validation – www.postman.com

 Log Monitoring & Performance Analysis – [Datadog](https://datadog.com) | [Sentry](https://sentry.io)

THANK YOU